

# OSLAB SKILL-8 PROGRAMS & OUTPUTS

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## Program-1 Code & Output:

```
sainath@sainath:~$ cd OSLAB
sainath@sainath:~/OSLAB$ ls
OS_PROJECT  filepermissions  practical_11  practical_4  practical_7  skill1  skill4  skill7
Skill2      fork             practical_12  practical_5  practical_8  skill2  skill5
exec        practical_10     practical_3   practical_6  practical_9  skill3  skill6
sainath@sainath:~/OSLAB$ mkdir skill8
sainath@sainath:~/OSLAB$ cd skill8
sainath@sainath:~/OSLAB/skill8$ gedit program1.c
sainath@sainath:~/OSLAB/skill8$ gcc program1.c -o program1
sainath@sainath:~/OSLAB/skill8$ cat program1.c
#include <stdio.h>
#include <stdlib.h>
#include <string.h>

int main() {
    char input[100], var[50], *value;

    printf("Enter variable: ");
    scanf("%49s", input);

    if (input[0] == '$') {
        strcpy(var, input + 1);
        value = getenv(var);

        if (value)
            printf("Value: %s\n", value);
        else
            printf("Undefined variable\n");
    } else {
        printf("Not a variable reference\n");
    }

    return 0;
}
sainath@sainath:~/OSLAB/skill8$ ./program1
Enter variable: $HOME
Value: /home/sainath
```

## Program-2 Code & Output:

```
sainath@sainath:~/OSLAB/skill8$ cat program2.c
#include <stdio.h>
#include <unistd.h>
#include <string.h>
#include <stdlib.h>

int main() {
    char command[20];

    printf("Mini Shell\n");

    while (1) {
        printf("> ");
        scanf("%19s", command);

        if (strcmp(command, "pwd") == 0) {
            char path[100];
            getcwd(path, sizeof(path));
            printf("%s\n", path);
        }
        else if (strcmp(command, "exit") == 0) {
            printf("Shell exited\n");
            break;
        }
        else {
            printf("Invalid command\n");
        }
    }

    return 0;
}

sainath@sainath:~/OSLAB/skill8$ ./program2
Mini Shell
> pwd
/home/sainath/OSLAB/skill8
> hello
Invalid command
> pwd
/home/sainath/OSLAB/skill8
> exit
Shell exited
```

## History:

```
655 cd OSLAB
656 ls
657 mkdir skill8
658 cd skill8
659 gedit program1.c
660 gcc program1.c -o program1
661 cat program1.c
662 ./program1
663 gedit program2.c
664 gcc program2.c -o program2
665 cat program2.c
666 ./program2
667 history
```